
Project Prometheus: A Work Plan and Strategic Assessment for AGI Benchmarking in Complex Games

Executive Summary

This document outlines a comprehensive work plan and strategic assessment for adapting and testing the Prometheus Proof-of-Concept (POC) AGI architecture within the domain of complex strategy games. The primary objective is to move the Prometheus demonstrator beyond its initial task of causal code refactoring and into dynamic, competitive environments that provide a robust testbed for its unique capabilities in causal reasoning, recursive self-improvement, and analogical thought.

The Prometheus architecture, with its emphasis on a **Causal Agentic Mesh (CAM)**, **Causal Reinforcement Learning from Self-Correction (CRLS)**, and a **Metacognitive Layer** capable of self-modification, is uniquely suited for games where success depends on understanding deep, underlying causal structures rather than mere correlational pattern matching.

Our assessment concludes that while classic board games like Chess offer valuable baselines, the most fertile grounds for demonstrating Prometheus's advanced capabilities lie in environments characterized by imperfect information, high-dimensional state spaces, and complex, non-obvious strategic layers. Consequently, this plan prioritizes **Go**, **No-Limit Texas Hold'em Poker**, and large-scale strategy "god games" such as **O A.D.** and **FreeCiv** as the ultimate benchmarks for the system.

The proposed work plan is structured in four phases, progressing from foundational integration in simple games to demonstrating advanced causal reasoning, mastering uncertainty, and finally achieving genuine recursive self-improvement in a complex, open-ended strategic world. This phased approach will systematically validate each component of the Prometheus architecture, aiming not merely to create a superhuman game-playing engine, but to produce a system capable of generating novel, explainable, and

causally-grounded strategies—a key milestone on the path to ultraintelligence.¹

1. Overview of the Prometheus POC Architecture

To select appropriate testbeds, it is essential to understand the core principles that differentiate the Prometheus architecture from conventional AI models.¹ The system is not designed for simple pattern recognition but for deep causal understanding and autonomous growth.

- **Core Differentiator: Causal Reasoning:** The POC's initial domain is causal code refactoring, where success is defined not by passing a unit test (correlational success) but by fundamentally improving a function's algorithmic efficiency (causal success).¹ This philosophy is central to the project. Prometheus is designed to build and maintain an **isomorphism**, or a structure-preserving map, between its internal models and the causal reality of its environment.¹
- Architectural Pillars ¹:
 1. **Causal Agentic Mesh (CAM):** A decentralized system of specialized, autonomous agents that collaborate to reason about problems. In a game context, these could be agents dedicated to economy, military tactics, or long-term strategy.
 2. **Causal Attention & CRLS:** A **Causal Attention Head** guides the system to focus on causally significant information, while **Causal Reinforcement Learning from Self-Correction (CRLS)** enables the system to learn from the causal correctness of its actions, not just the binary win/loss outcome.
 3. **Recursive Self-Improvement (RSI) via a Metacognitive Layer:** The architecture features a distinct **Metacognitive Layer** (the "mind") containing a **Self-Modification Module (SMM)** and an **Introspection and Evaluation Engine (IEE)**. This layer observes and analyzes the performance of the **Capability Layer** (the "body") and can rewrite its own code to improve its performance. This process is conceptualized as a Hofstadterian "Strange Loop," where the system reasons about and improves its own reasoning processes.¹
 4. **Internal Governance:** A **Modern Centrencephalic System (MCS)** acts as an internal alignment governor, ensuring the RSI process does not lead to goal drift and that the agent's behavior remains aligned with its core principles.¹

2. Assessment of Game Environments

The suitability of a game as a benchmark for Prometheus is determined by its capacity to test these core architectural principles. We evaluated a range of game genres based on criteria derived from the Prometheus design documents.

Evaluation Criteria:

1. **Causal Complexity:** The depth and non-obviousness of the game's underlying causal structure.
2. **Causal vs. Correlational Success:** The degree to which lucky or brute-force wins can be distinguished from strategically sound (causally correct) play.
3. **Potential for Analogy & Abstraction:** The richness of high-level concepts, patterns, and transferable strategies.
4. **Imperfect Information:** The requirement to reason about hidden states and opponent intentions.
5. **Technical Feasibility:** The availability of stable, programmatic APIs for agent interaction.

Assessment Summary

Game Category	Representative Games	Causal Complexity	Causal vs. Corr.	Analogy Potential	Imperfect Info	Tech. Feasibility	Suitability for Prometheus
Simple Abstract	Connect4, Draughts	Low	Low	Low	None	High	Low (Baseline only)
Stochastic Abstract	Backgammon	Medium	Medium	Medium	Low (Dice)	High	Medium (Good for CRLS)
Complex Abstract	Chess, Go	High	High	High	None	High	Very High (Excellent for Causal

							Attention & Analogy)
Imperfect Info (Cards)	Poker, Bridge	Very High	Very High	High	Very High	Medium-High	Excellent (Prime test for reasoning under uncertainty)
4X / God Games	FreeCiv, O A.D.	Excellent	High	Excellent	High (Fog of War)	High	Excellent (Ultimate test for RSI & CAM)
RTS (Simplified)	MicroRTS	Medium	Medium	Medium	High (Fog of War)	High	High (Good for real-time CAM testing)

Analysis:

- Classic Games (Checkers, Chess, Go):** AI has a long history with these games.² While checkers is largely solved⁵, Chess and especially Go remain profound strategic challenges. For Prometheus, the goal would not be to merely win through superior search, but to develop a causal understanding of positional concepts (e.g., "pawn structure" in chess, "influence" in Go). Go is a particularly strong candidate due to its emergent properties and the difficulty of creating a simple evaluation function, making it a prime test for the Causal Attention Head.⁶

- **Card Games (Poker, Bridge):** These games are premier environments for testing reasoning under uncertainty. Poker, a game of imperfect information, requires sophisticated opponent modeling and probabilistic reasoning, making it an ideal domain to test the CAM and CRLS.⁷ The distinction between a causally "correct" play (positive expected value) and a lucky outcome is explicit. Bridge adds a layer of cooperative communication through bidding, providing a testbed for multi-agent coordination within the CAM.¹⁰
- **God Games (FreeCiv, O A.D., MicroRTS):** These environments represent the ultimate challenge. Their vast state spaces, long-term planning requirements, and complex interplay of economic, diplomatic, and military systems make them ideal for testing the full Prometheus architecture. The open-source nature of games like O A.D. and FreeCiv, complete with existing AI APIs, provides high technical feasibility.¹² These games offer a clear domain for the SMM to engage in true Recursive Self-Improvement, modifying its own strategic source code to invent novel approaches to winning—the core thesis of the "intelligence explosion".¹

3. Detailed Work Plan

This work plan adopts a progressive, four-phase approach, systematically increasing complexity to validate the Prometheus architecture.

Phase 1: Foundational Integration & Baseline Testing (Months 1-6)

Objective: Adapt the Prometheus POC for game-playing and establish baseline functionality in simple, well-understood environments.

- **Task 1.1: Domain Abstraction Layer Development:** Engineer a universal interface that translates diverse game states (boards, hands, maps) into a standardized format for the Capability Layer. This task directly implements the Hofstadterian concept of creating an "isomorphism" between the game world and the agent's internal model.¹
- **Task 1.2: Baseline Testing in Solved/Simple Games:**
 - **Environment:** Draughts (Checkers) and Backgammon.
 - **Methodology:** Implement the core RSI loop to validate its functionality. The CAM will be minimal (e.g., "Offense" and "Defense" agents). The IEE will use a simple win/loss fitness function. Performance will be measured against established benchmarks like Arthur Samuel's checkers program and TD-Gammon.²

- **Task 1.3: API Integration for Advanced Environments:** Establish and verify robust, programmatic control interfaces for the primary target games:
 - Chess: Universal Chess Interface (UCI).¹⁵
 - Go: Go Text Protocol (GTP).¹⁸
 - O A.D.: Utilize the built-in JavaScript scripting API for initial tests and the external Reinforcement Learning (RL) HTTP interface for advanced implementation.¹²
 - FreeCiv: Leverage the C-based server and existing AI modules for integration.¹⁴

Phase 2: Causal Reasoning in Perfect Information Games (Months 7-18)

Objective: Demonstrate that Prometheus can achieve elite performance in complex, perfect-information games through superior causal understanding, rather than brute-force search alone.

- **Task 2.1: Mastering Chess through Causal Concepts:**
 - **Methodology:** The CAM will be configured with agents specializing in abstract chess concepts like "King Safety," "Pawn Structure," and "Piece Activity." The Causal Attention Head will be trained to identify causally critical squares and pieces. The IEE's fitness function will reward improvements in these positional metrics, guiding the SMM to refine the agent's understanding of their relative importance.
 - **Success Metric:** Achieve Grandmaster-level play while utilizing a demonstrably smaller search tree than leading engines like Stockfish.¹⁵
- **Task 2.2: Mastering Go through Emergent Properties:**
 - **Methodology:** Apply a similar approach to Go, with CAM agents focused on "Territory," "Influence," and "Life-and-Death" problems. The vast and abstract nature of Go makes it a crucial test for the Causal Attention Head's ability to discard spurious correlations.
 - **Success Metric:** Match the performance of AlphaGo/AlphaZero while providing an explainable, causal model for its strategic decisions, moving beyond the "black box" nature of pure deep learning approaches.⁶

Phase 3: Reasoning Under Uncertainty in Imperfect Information Games (Months 19-30)

Objective: Demonstrate mastery of games requiring reasoning about hidden states,

opponent modeling, and probabilistic outcomes.

- **Task 3.1: Achieving Expert Play in No-Limit Texas Hold'em:**

- **Methodology:** This is a primary testbed. The CAM will feature agents for "Hand Strength Estimation," "Opponent Modeling" (learning opponent tendencies), and "Strategic Deception" (bluffing/trapping). The CRLS loop is essential for learning from outcomes where hidden information is revealed.
- **Success Metric:** Consistently defeat professional human players and state-of-the-art poker AIs like DeepStack.⁸ A key qualitative metric will be the agent's ability to generate a causal explanation for its actions (e.g., "I executed a check-raise because my opponent model indicated a high probability they were continuation-betting with a wide range").

- **Task 3.2: Cooperative Reasoning in Contract Bridge:**

- **Methodology:** Focus on the bidding phase as a complex, cooperative language game. The CAM must learn the causal links between bids and hand information to build a coherent model of its partner's hidden hand.
- **Success Metric:** Achieve world-class performance in competitive play, demonstrating robust partnership bidding and card play.

Phase 4: Large-Scale Strategic Planning and the "Strange Loop" (Months 31-48)

Objective: Tackle a complex real-time or turn-based "god game" to test long-term planning and the full potential of the Recursive Self-Improvement cycle.

- **Task 4.1: Demonstrating Emergent Strategy in 0 A.D. or FreeCiv:**

- **Methodology:** Deploy the complete Prometheus architecture. The CAM will be at its most complex, with high-level agents for Economy, Research, Diplomacy, and Military Command, each potentially composed of further sub-agents. The Metacognitive Layer will be granted maximum autonomy. The SMM will be empowered to propose and test modifications to the agent's own strategic source code—for instance, altering its economic build orders, technology priorities, or diplomatic stances based on IEE feedback. This operationalizes the "Strange Loop," where the agent actively improves its own methods for improvement.¹
- **Success Metric:** Defeat the most difficult built-in AIs and expert human players. The ultimate demonstration of success will be the generation of a verifiably novel and effective strategy, proving that the system has transcended its initial programming and entered a genuine state of creative self-improvement.

4. Assessment and Verification Protocol

Success will be measured by a combination of quantitative performance metrics and qualitative analysis of the agent's reasoning processes.

- **Performance Metrics:** Win/loss rates against established AI benchmarks, Elo ratings against human players, and task-specific metrics (e.g., expected value calculations in poker, economic output in O A.D.).
 - **Causal Analysis:** Throughout all phases, the agent's internal "deliberations" will be logged. Success requires not just winning, but demonstrating that wins are correlated with improvements in the agent's causal models. For example, in Phase 2, we must show that a higher win rate in chess is causally linked to a better understanding of "King Safety," as measured by the IEE.
 - **Alignment and Stability:** The Modern Centrencephalic System (MCS) will continuously monitor the agent's behavior. Its role is to prevent "reward hacking" (e.g., finding an engine exploit instead of learning strategy) and ensure the RSI process does not cause catastrophic goal drift, thereby maintaining alignment with the project's core objectives.¹
 - **Novelty Assessment:** In Phase 4, any strategies generated by the SMM will be rigorously analyzed by human domain experts to verify their novelty and effectiveness. The generation of a truly new and powerful game strategy would serve as the ultimate validation of the Prometheus project.
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